DON BOSCO COLLEGE(CO-ED)

B.SC COMPUTER SCIENCE DEPARTMENT

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COLLEGE CODE:358

PROJECRT TITLE: CRICKET PLAYER PERFORMANCE TRACKING APPLICATION :CRICKETER

TEAM MEMBERS NAME:

1.PRAVEEN S	(35823U18033)
2.PRAVEEN T	(35823U18034)
3.PRICILA MARY A	(35823U18035)
4.PRICILLA MARY MA	(35823U18036)
5.PRINCY JUDITH	(35823U18037)

FACULTY NAME:Ms.POOVARASI

ABSTRACT

Cricket is a sport that thrives on precision, statistics, and continuous improvement. Every run scored, wicket taken, and catch held contributes to the outcome of a match, and the ability to track these details accurately is vital for coaches, analysts, and players. Traditionally, performance tracking in cricket has relied on manual scorecards, spreadsheets, or fragmented systems that fail to provide a holistic view of player development. These methods often lead to errors, inconsistencies, and delays in generating meaningful insights. In the modern era of sports analytics, where data-driven decisions can determine the difference between victory and defeat, there is a pressing need for a centralized, automated solution.

The Cricket Player Performance Tracking Application is designed to meet this need by offering a comprehensive platform that integrates match statistics, training progress, and fitness monitoring into a single system. The application records batting, bowling, and fielding data, calculates averages and strike rates automatically, and stores fitness metrics to evaluate player readiness. It introduces validation rules to ensure accuracy in data entry, preventing incorrect or incomplete records. Record types and customized layouts distinguish between different formats of the game such as Test, ODI, and T20, allowing structured categorization of performance data.

Automation is a key feature of the system. Formula fields compute batting averages, strike rates, and bowling economy rates without manual intervention, reducing human error and saving time. Scheduled workflows generate player performance reports, notify coaches of training schedules, and update dashboards with real-time insights. Interactive dashboards provide visual representations of player form, team strengths, and match outcomes, enabling coaches and managers to make informed decisions.

Being cloud-based, the application ensures secure storage, scalability, and accessibility from anywhere at any time. It empowers coaches to design personalized training programs, helps managers in team selection and strategy formulation, and provides players with transparent feedback on their performance. Ultimately, the Cricket Player Performance Tracking Application bridges the gap between traditional scorekeeping and advanced sports analytics, offering a data-driven approach to player development and team management. This project demonstrates how technology can transform cricket performance evaluation, making it more efficient, transparent, and impactful for the long-term growth of the sport.

INTRODUCTION

Cricket is one of the most widely played and celebrated sports across the world, particularly in countries such as India, Australia, England, and South Africa. It is a game that thrives on precision, teamwork, and strategy, where every run scored, wicket taken, or catch held can change the course of a match. With the increasing professionalization of the sport, performance tracking has become a critical component in evaluating players and enhancing team outcomes. Traditionally, cricket statistics such as batting averages, strike rates, bowling economy rates, and fitness levels were recorded manually using scorecards or spreadsheets. While these methods provided basic information, they often lacked depth, accuracy, and the ability to generate real-time insights. As a result, coaches and managers struggled to make data-driven decisions, and players had limited access to personalized feedback on their performance.

The Cricket Player Performance Tracking Application is designed to address these challenges by offering a centralized digital platform that integrates match statistics, training progress, and fitness monitoring. The system provides coaches, analysts, and players with a structured environment to record and access detailed performance data. Unlike conventional scorekeeping, the application transforms raw numbers into actionable insights, highlighting strengths, weaknesses, and consistency levels. This ensures that performance evaluation is not only accurate but also meaningful in guiding player development and team strategy.

The application introduces several innovative features. Custom modules are created to manage player profiles, match records, and training sessions. Automated formula fields calculate batting averages, strike rates, and bowling economy rates without manual intervention, reducing human error and saving time. Validation rules are implemented to ensure accurate data entry, preventing incorrect or incomplete records. Record types and customized layouts distinguish between different formats of the game such as Test, ODI, and T20, allowing structured categorization of performance data. Interactive dashboards and reports provide real-time insights into player form, team strengths, and match outcomes, enabling coaches and managers to make informed decisions quickly.

Beyond match statistics, the application also emphasizes player development through fitness monitoring and training progress tracking. By recording fitness scores and training schedules, the system helps coaches design personalized training programs tailored to each player's

needs. Automated notifications remind players of upcoming training sessions and performance reviews, ensuring consistent engagement. This holistic approach ensures that both physical fitness and skill development are monitored alongside match performance.

In essence, the Cricket Player Performance Tracking Application represents a step forward in the digital transformation of sports management. It bridges the gap between traditional scorekeeping and advanced analytics, empowering teams to harness data for continuous improvement. By combining technology with the passion for cricket, the application contributes to the professional growth of players, enhances team management efficiency, and supports the overall development of the sport.

PROBLEM STATEMENT

Cricket is a sport that depends heavily on accurate performance evaluation, consistent training, and strategic decision-making. However, most teams and academies still rely on manual methods such as paper scorecards, registers, or basic spreadsheets to record player statistics. These traditional approaches are prone to errors, inconsistencies, and delays, making it difficult to generate timely insights into player development. As the game becomes increasingly competitive and data-driven, the absence of a unified and automated system creates significant challenges for coaches, managers, and players.

Problems Faced by Coaches

Coaches are responsible for monitoring player progress and designing training programs. Yet, without an automated system, they must calculate batting averages, strike rates, and bowling economy rates manually. This process is time-consuming and often inaccurate. Fitness records and training schedules are also updated manually, which limits the ability to track improvements over time. As a result, coaches struggle to provide personalized feedback and targeted training interventions.

Problems Faced by Team Managers

Managers require accurate and timely reports to evaluate team strengths, player form, and match outcomes. However, compiling performance data from scattered registers and spreadsheets is labor-intensive and often delayed. Without real-time insights, managers cannot make quick, data-driven decisions regarding team selection or match strategy. This lack of integration reduces overall efficiency and weakens the team’s competitive edge.

Problems Faced by Players

Players expect transparent feedback and structured guidance to improve their performance. In the absence of an automated system, they do not receive personalized performance reports or fitness progress updates. Training improvements are not tracked systematically, and players often remain unaware of their strengths and weaknesses. This lack of engagement limits motivation and hinders long-term development.

Overall Challenge

The absence of a centralized performance tracking application results in:

- Inaccurate evaluations due to manual calculations
- Disorganized and scattered performance data
- Lack of instant reporting and analytics
- Limited player development strategies
- Inefficient managerial decision-making

These limitations negatively affect service quality, operational efficiency, and long-term scalability in cricket management. Therefore, there is a pressing need for a secure, automated, and centralized system that can streamline performance tracking, enhance data accuracy, and strengthen player development strategies.

PROPOSED SOLUTION

To overcome the challenges of manual record-keeping, scattered data, and delayed reporting in cricket management, the Cricket Player Performance Tracking Application is proposed as a centralized, automated, and intelligent platform. This solution integrates match statistics, training progress, and fitness monitoring into a single system, ensuring accuracy, efficiency, and transparency in performance evaluation.

Centralized Data Management

The application introduces structured modules to organize information effectively. Player Profiles store personal details, career statistics, and fitness records. Match Records capture batting, bowling, and fielding data, while Training Session Logs track fitness drills and practice schedules. By consolidating all data into a unified database, the system eliminates duplication and improves accessibility for coaches, managers, and players.

Automation of Performance Calculations

Manual calculations of batting averages, strike rates, and bowling economy rates are replaced with automated formula fields. Fitness scores are computed based on training inputs, ensuring precise evaluation of player readiness. This automation reduces human error, saves time, and provides instant insights into player performance.

Data Validation Controls

To maintain high data integrity, validation rules are implemented. These rules restrict invalid entries such as incorrect match statistics, incomplete fitness metrics, or unrealistic training data. Standardized formats ensure consistency across records, improving reliability and usability of the system.

Record Types and User Interface Customization

Different formats of cricket—Test, ODI, and T20—require distinct performance tracking. The application introduces record types for each format, with customized layouts to categorize data appropriately. This structured approach enhances usability and ensures that performance evaluation is tailored to the specific demands of each format.

Reporting and Analytical Dashboards

The system generates dynamic reports and interactive dashboards that provide real-time insights into player form, team strengths, and match outcomes. Coaches can access individual player performance reports, managers can view team comparison reports, and players can track their fitness progress. These analytical tools support strategic decision-making and long-term player development.

Automation of Notifications and Workflows

The application includes automated workflows to handle routine tasks. Scheduled notifications remind players of training sessions, while automated reports are generated after matches to provide instant feedback. This ensures consistent engagement and reduces administrative workload for coaches and managers.

WORKFLOW DIAGRAM

Step 1: Data Entry

- Match officials, coaches, or analysts enter player statistics such as runs scored, balls faced, wickets taken, overs bowled, and catches.
- Training staff input fitness metrics and practice session details.
- Players may also update personal fitness logs or training attendance.

Step 2: Data Validation

- Validation rules check the accuracy of entries (e.g., ensuring overs bowled do not exceed match limits, strike rates are calculated correctly, and fitness scores are within realistic ranges).
- Incorrect or incomplete data is flagged for correction before being stored.

Step 3: Automated Calculations

- Formula fields compute batting averages, strike rates, bowling economy rates, and fitness scores automatically.
- This reduces manual workload and ensures precision in performance evaluation.

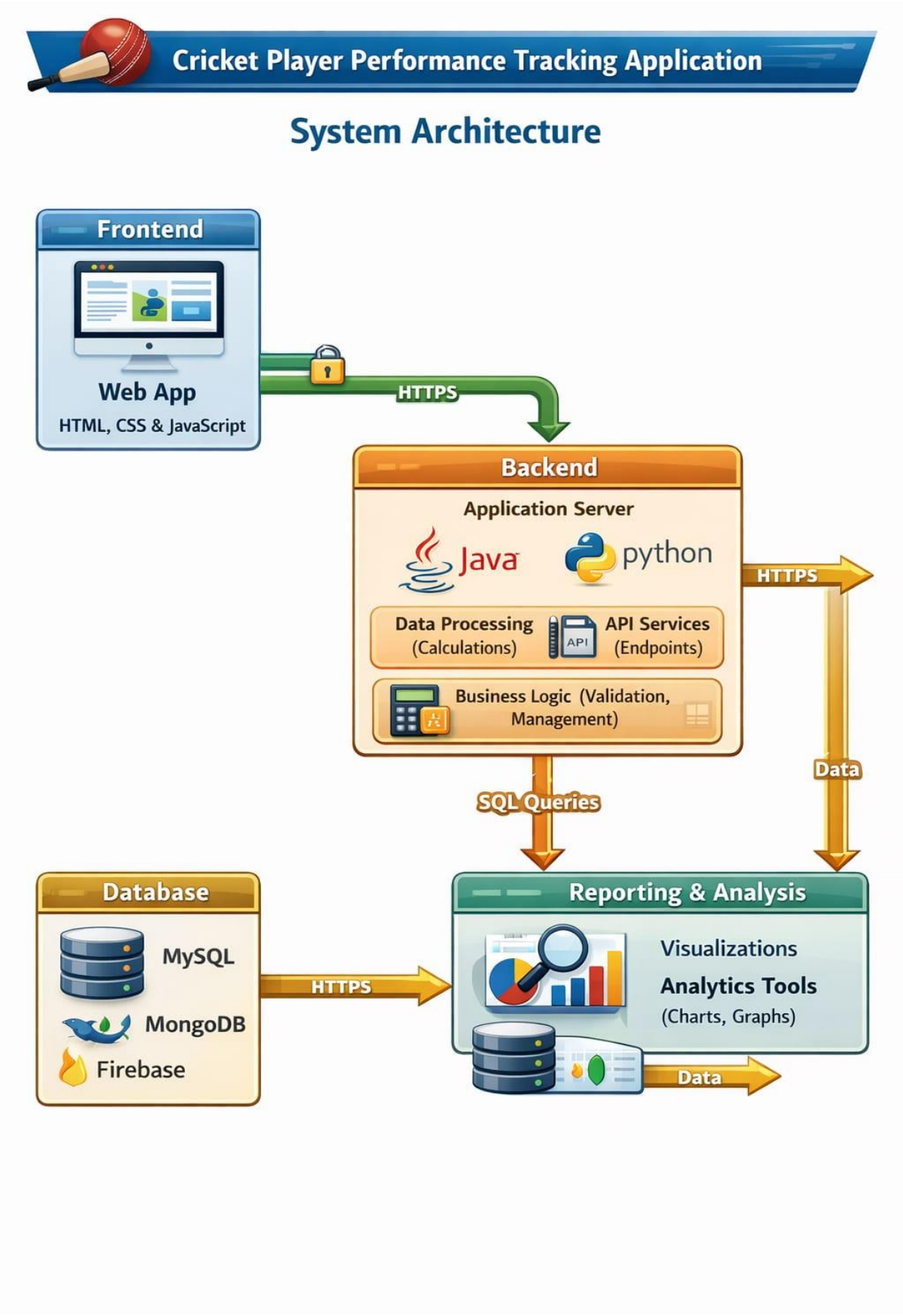
Step 4: Centralized Data Storage

- All validated and processed data is stored in a unified database.
- Player Profiles, Match Records, and Training Sessions are linked, ensuring easy retrieval and structured organization.

Step 5: Reporting & Dashboards

- The system generates real-time reports such as individual player performance, team comparisons, and fitness progress.
- Interactive dashboards visualize trends in batting form, bowling consistency, and overall team strengths.

SYSTEM ARCHITECTURE



Key Components

1. User Layer

- Players: Input fitness data, training attendance, and receive personalized feedback.
- Coaches: Enter match statistics, monitor training progress, and access performance reports.
- Managers/Analysts: Generate team-level reports, compare players, and make strategic decisions.

2. Application Layer

- Data Entry Module: Interfaces for recording match statistics, training sessions, and fitness metrics.
- Validation Engine: Ensures accuracy of inputs (e.g., correct overs bowled, realistic fitness scores).
- Calculation Engine: Automated formula fields compute batting averages, strike rates, bowling economy, and fitness scores.
- Workflow Automation: Scheduled tasks trigger notifications, generate reports, and update dashboards.

3. Database Layer

- Player Profile Table: Stores personal details, career stats, and fitness records.
- Match Record Table: Logs batting, bowling, and fielding statistics for each match.
- Training Session Table: Tracks fitness drills, practice schedules, and progress.
- Performance Report Table: Consolidates calculated metrics for analysis and visualization.

4. Analytics Layer

- Reports Module: Generates detailed performance reports for individuals and teams.
- Dashboards Module: Provides interactive visualizations of batting form, bowling consistency, and fitness progress.
- Comparative Analysis: Enables side-by-side evaluation of players across formats (Test, ODI, T20).

5. **Cloud Infrastructure**

- Ensures secure data storage, scalability, and 24/7 accessibility.
- Supports integration with external devices (e.g., fitness trackers, video analysis tools).

SUMMARY

The project titled "Cricket Player Performance Tracking Application" presents the design and implementation of a centralized digital system to streamline player evaluation and team management.

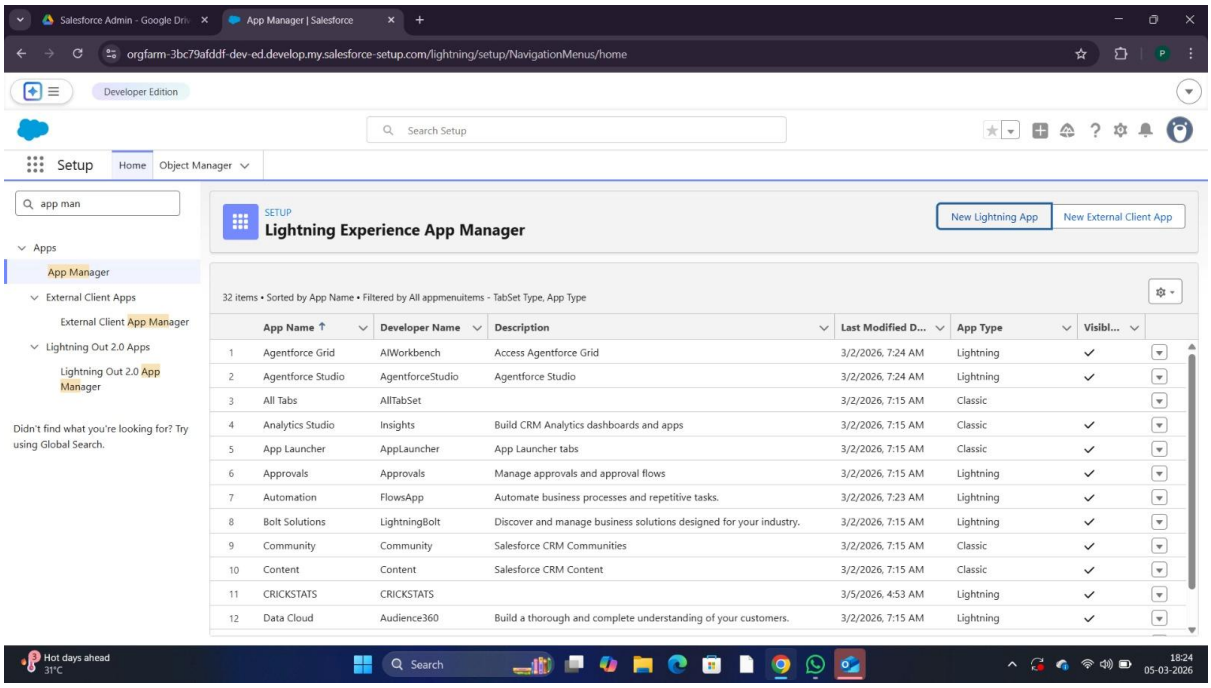
The application replaces traditional manual scorekeeping and fragmented record-keeping with an integrated and automated solution that ensures accuracy, efficiency, and transparency in performance tracking.

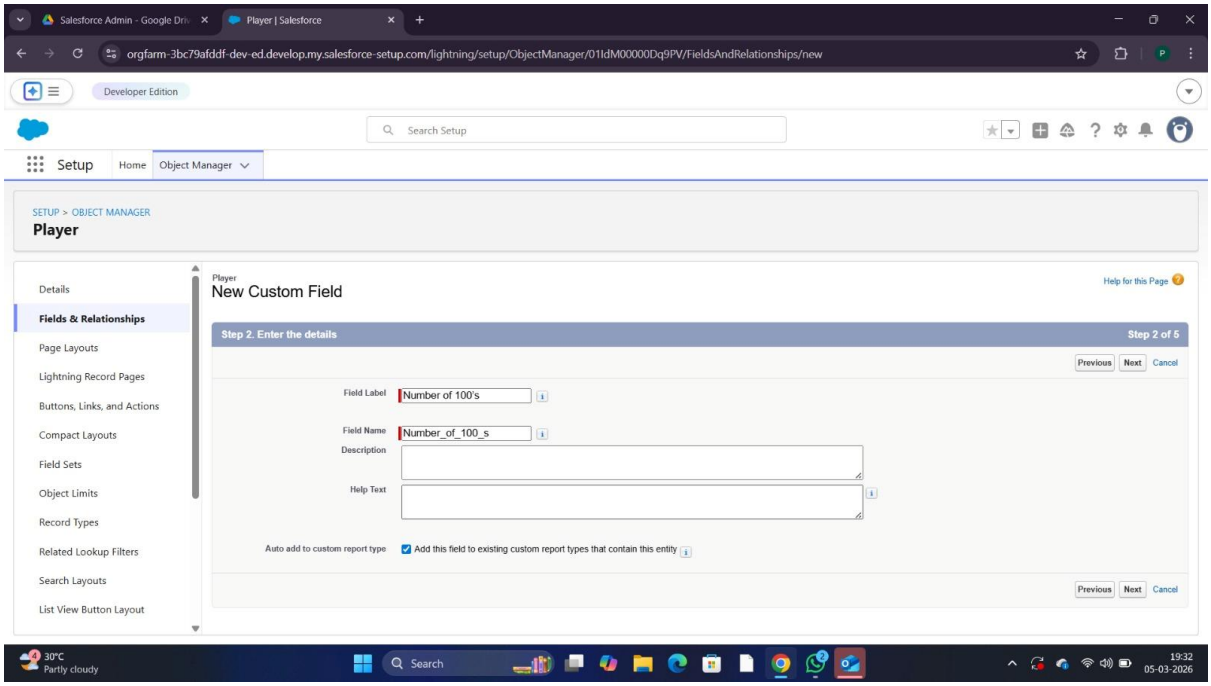
Custom modules such as Player Profiles, Match Records, and Training Sessions are developed to organize and manage data in a structured manner. Automated formula fields calculate batting averages, strike rates, bowling economy rates, and fitness scores, thereby minimizing human errors and reducing manual workload. Validation rules enforce correct formats for match statistics and training inputs, ensuring consistency and reliability of data. Record types are configured to distinguish between different formats of cricket—Test, ODI, and T20—allowing better categorization and analysis of performance data.

The system also incorporates workflow automation, enabling scheduled notifications for training sessions, automated generation of performance reports, and real-time updates to dashboards. Comprehensive reports and interactive dashboards deliver insights into player form, team strengths, and match outcomes. These analytical tools empower coaches to design personalized training programs, managers to make strategic decisions, and players to track their progress transparently.

Being cloud-based, the application guarantees secure data storage, scalability, and accessibility from anywhere at any time. Its architecture supports future enhancements such as integration with wearable fitness devices, AI-driven performance predictions, and video analytics. By reducing repetitive manual tasks and providing real-time insights, the system enhances collaboration between coaches, managers, and players.

SCREENSHOT

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Step 2. Enter the details
Step 2 of 5
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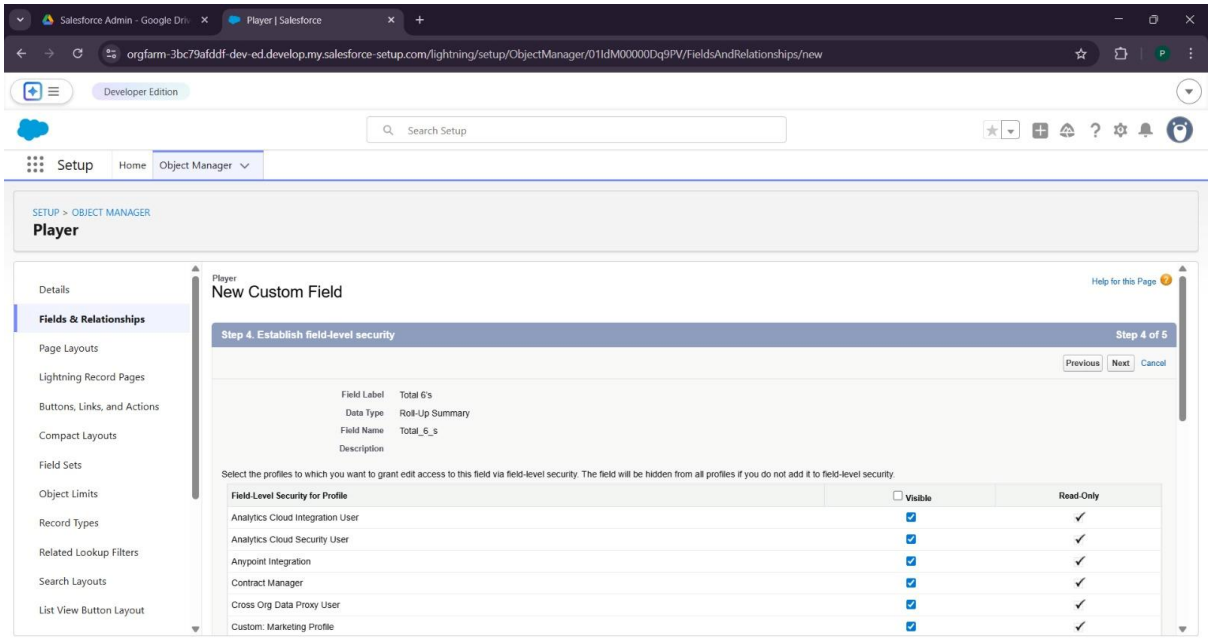
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Step 4 of 5
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Total 6's
Data Type
Roll-Up Summary
Field Name
Total_6_s
Description
Select the profiles to which you want to grant edit access to this field via field-level security. The field will be hidden from all profiles if you do not add it to field-level security.
Field-Level Security for Profile
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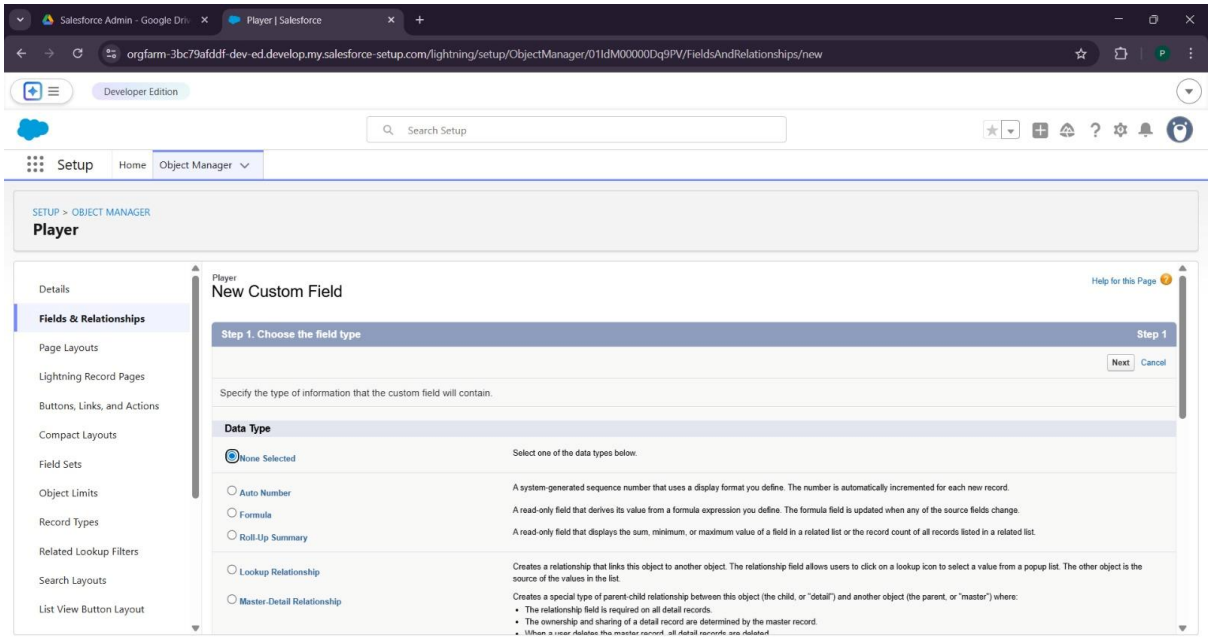
Read-Only
Analytics Cloud Integration User
Analytics Cloud Security User
Anypoint Integration
Contract Manager
Cross Org Data Proxy User
Custom: Marketing Profile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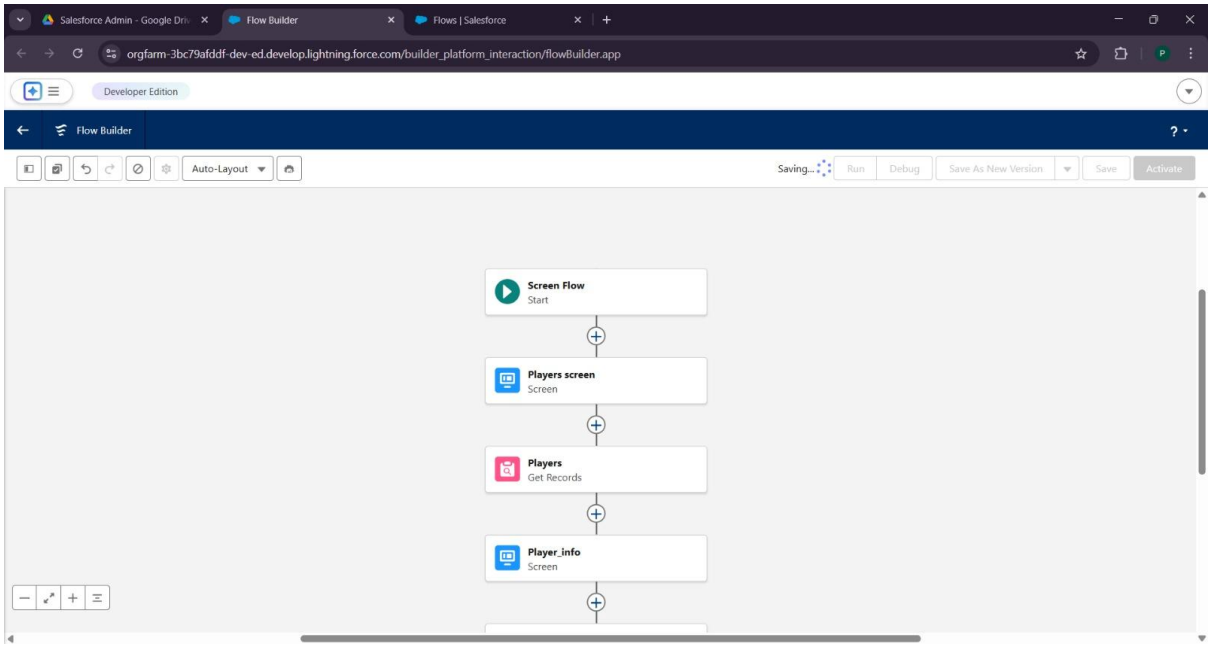
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts
List View Button Layout
Player
New Custom Field
Help for this Page
Step 1. Choose the field type
Step 1
Next
Cancel
Specify the type of information that the custom field will contain.
Data Type
None Selected
Select one of the data types below.
Auto Number
A system-generated sequence number that uses a display format you define. The number is automatically incremented for each new record.
Formula
A read-only field that derives its value from a formula expression you define. The formula field is updated when any of the source fields change.
Roll-Up Summary
A read-only field that displays the sum, minimum, or maximum value of a field in a related list or the record count of all records listed in a related list.
Lookup Relationship
Creates a relationship that links this object to another object. The relationship field allows users to click on a lookup icon to select a value from a popup list. The other object is the source of the values in the list.
Master-Detail Relationship
Creates a special type of parent-child relationship between this object (the child, or "detail") and another object (the parent, or "master") where:
The relationship field is required on all detail records.
The ownership and sharing of a detail record are determined by the master record.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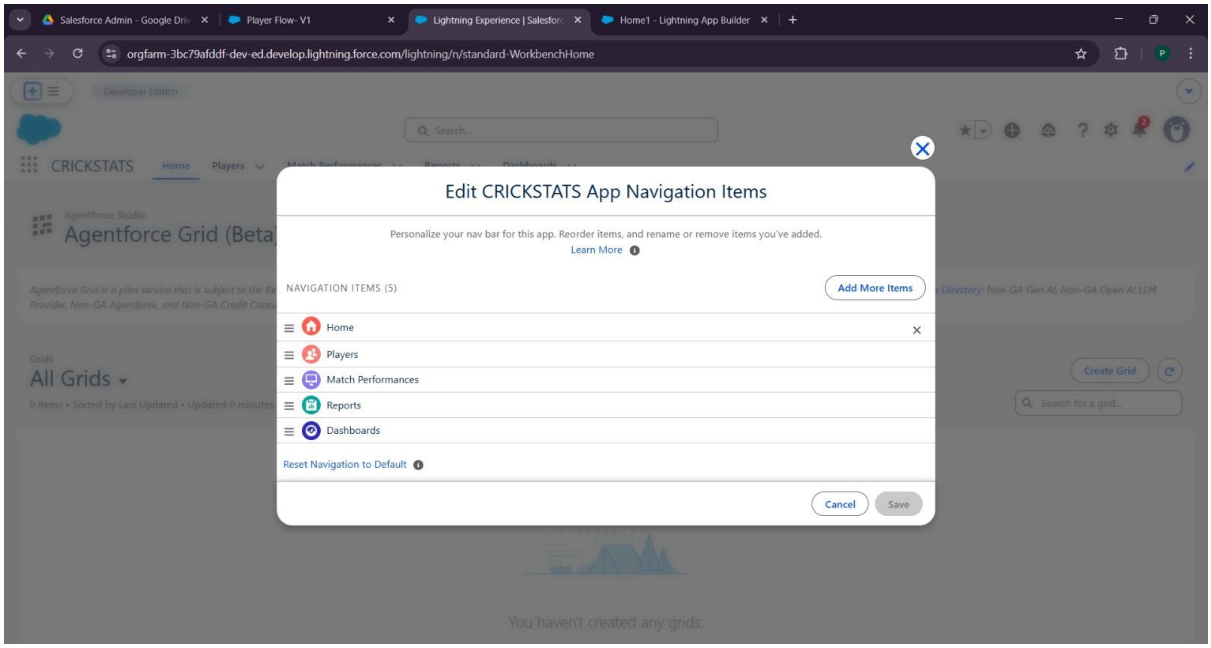
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Grids
All Grids
Create Grid
Search for a grid...
You haven't created any grids.
Edit CRICKSTATS App Navigation Items
Personalize your nav bar for this app. Reorder items, and rename or remove items you've added.
Learn More
NAVIGATION ITEMS (5)
Add More Items
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Match Performances
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Dashboards
Reset Navigation to Default
Cancel
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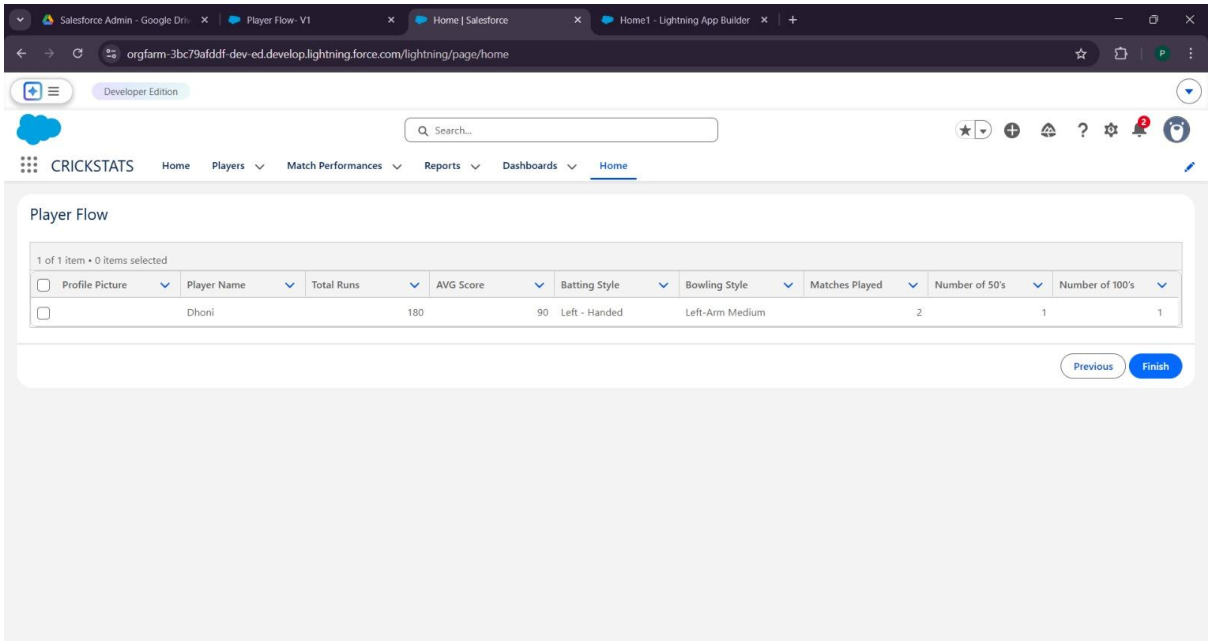
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Player Flow
1 of 1 item • 0 items selected
Profile Picture
Player Name
Total Runs
AVG Score
Batting Style
Bowling Style
Matches Played
Number of 50's
Number of 100's
Dhoni
180
90
Left - Handed
Left-Arm Medium
2
1
1
Previous
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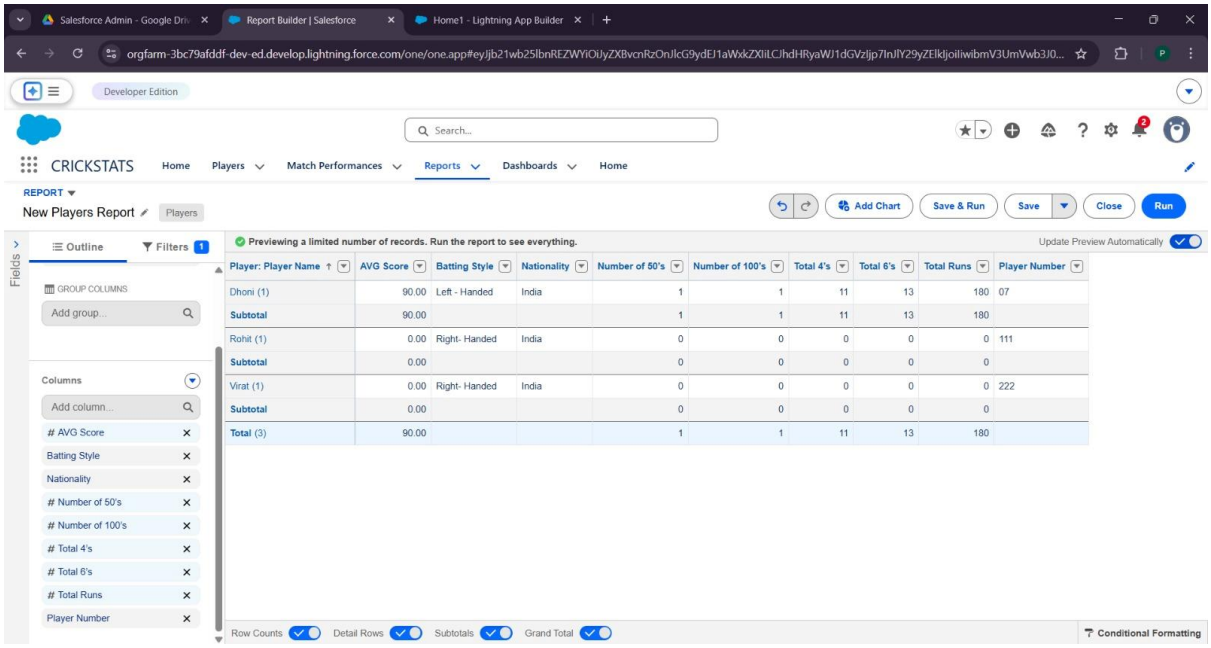
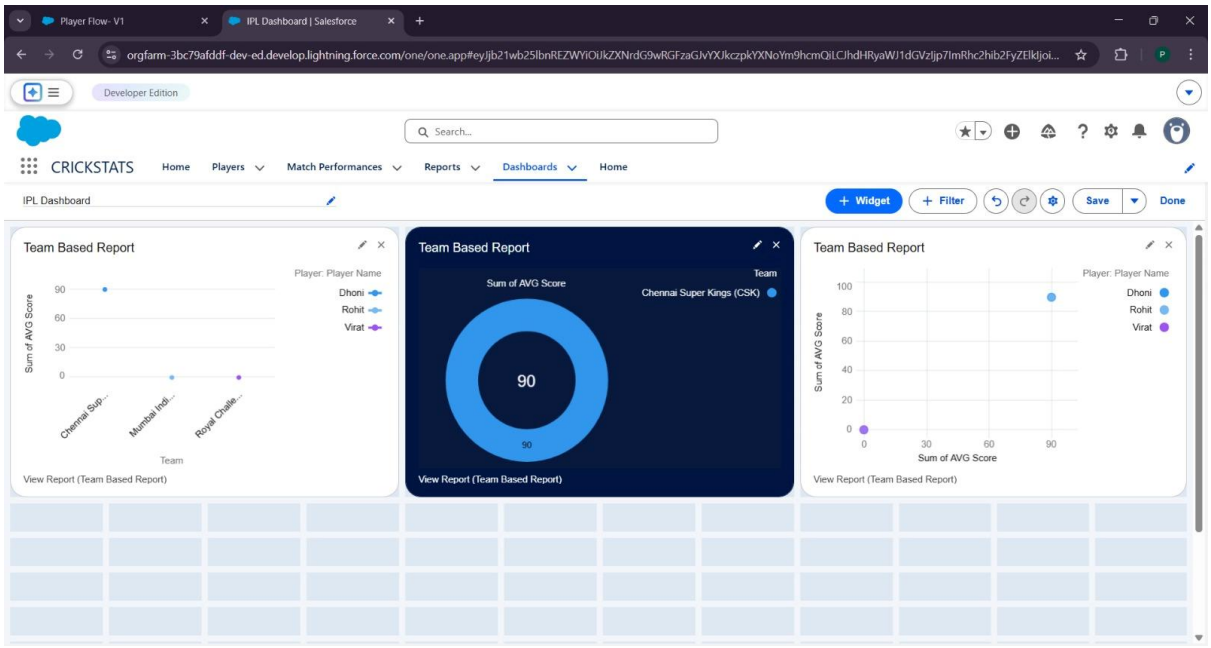


Player: Player Name	AVG Score	Batting Style	Nationality	Number of 50's	Number of 100's	Total 4's	Total 6's	Total Runs	Player Number
Dhoni (1)	90.00	Left - Handed	India	1	1	11	13	180	07
Subtotal	90.00			1	1	11	13	180	
Rohit (1)	0.00	Right- Handed	India	0	0	0	0	0	111
Subtotal	0.00			0	0	0	0	0	
Virat (1)	0.00	Right- Handed	India	0	0	0	0	0	222
Subtotal	0.00			0	0	0	0	0	
Total (3)	90.00			1	1	11	13	180	



CONCLUSION

The Cricket Player Performance Tracking Application successfully demonstrates how technology can transform the way player statistics, fitness progress, and training outcomes are managed in modern cricket. By replacing traditional manual scorekeeping and fragmented record-keeping with a centralized, automated system, the application addresses critical challenges such as inaccurate calculations, scattered data, and delayed reporting.

Through the use of structured modules like Player Profiles, Match Records, and Training Sessions, the system ensures that all performance data is organized and easily accessible. Automated formula fields reduce human error by calculating batting averages, strike rates, and bowling economy rates instantly, while validation rules enforce accuracy and consistency in data entry. Record types tailored to different formats of the game—Test, ODI, and T20—allow for precise categorization and analysis, ensuring that performance evaluation is relevant to the specific demands of each format.

The inclusion of workflow automation further enhances efficiency by generating scheduled reports, sending training reminders, and updating dashboards in real time. Interactive dashboards and analytical reports provide coaches, managers, and players with actionable insights into player form, team strengths, and match outcomes. This empowers stakeholders to make informed, data-driven decisions that directly impact team performance and player development.

Being cloud-based, the application guarantees secure data storage, scalability, and accessibility from anywhere, making it suitable for both professional teams and training academies. The architecture also supports future enhancements such as integration with wearable fitness devices, AI-driven performance predictions, and video analytics, ensuring that the system remains adaptable to evolving technological trends in sports management.

In conclusion, the Cricket Player Performance Tracking Application represents a significant step forward in the digital transformation of cricket. It bridges the gap between traditional scorekeeping and advanced analytics, strengthens collaboration between coaches, managers, and players, and supports long-term player growth. By combining automation, structured data management, and real-time insights, the application not only improves efficiency but also contributes to the overall competitiveness and sustainability of cricket as a sport.